

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		52451	
Total Threats Mitigated	27	Total Packets Analyzed		11290912202	
High Priority	0	Malicious Packets		11280229098	
Low Priority	27	True Negatives Passed		43301	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		18	
False Positives	77	UDP Flood		0	
FP Rate	0.14%	Uncertain		9	
IF Precision	99.36%				
IF Recall	94.53%	Action		Count	
IF F1-Score	96.88%	Quarantined		5	
IF Accuracy	98.89%	Time Ban		4	
		Blackhole		4	
		Released		6	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.36%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.53%	Share of actual attacks successfully flagged
F1-Score	96.88%	Balanced measure combining Precision and Recall
Accuracy	98.89%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.14%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.47%	Actual attacks that went undetected
True Positive Rate (TPR)	94.53%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.86%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 9048	FN 524
Actual: Normal	FP 58	TN 42777

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	9048	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Response Latency

Metric	Value	Description
Detection Time	300.8 ms	Average time taken to detect an attack
Mitigation Response Time	5.0 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.75%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	15.85%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	15.13%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	8	4	2026-09-14 10:12:15	2026-09-14 10:28:45
10.0.0.17	8	3	2026-09-14 10:12:16	2026-09-14 10:28:45
10.0.0.9	2	3	2026-09-14 10:09:05	2026-09-14 10:09:59
10.0.0.18	7	2	2026-09-14 10:12:16	2026-09-14 10:28:45
10.0.0.19	7	2	2026-09-14 10:12:20	2026-09-14 10:28:48
10.0.0.5	3	1	2026-09-14 10:29:02	2026-09-14 10:30:40
10.0.0.1	1	1	2026-09-14 10:29:07	2026-09-14 10:29:18
10.0.0.8	1	1	2026-09-14 10:29:07	2026-09-14 10:29:18

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-14 10:09:05	10.0.0.9	DDoS	Uncertain	69.9%	Low	Quarantined
2026-09-14 10:09:16	10.0.0.9	DDoS	Uncertain	69.9%	Low	Released
2026-09-14 10:12:15	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 10:12:16	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 10:12:16	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 10:12:20	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 10:12:26	10.0.0.16	DDoS	SYN Flood	93.4%	Critical	Time Ban (1m)
2026-09-14 10:12:26	10.0.0.18	DDoS	SYN Flood	91.0%	Critical	Time Ban (1m)
2026-09-14 10:12:26	10.0.0.18	DDoS	SYN Flood	91.0%	Critical	Time Ban
2026-09-14 10:12:26	10.0.0.16	DDoS	SYN Flood	93.4%	Critical	Time Ban
2026-09-14 10:12:27	10.0.0.17	DDoS	SYN Flood	91.0%	Critical	Time Ban (1m)
2026-09-14 10:12:27	10.0.0.17	DDoS	SYN Flood	91.0%	Critical	Time Ban
2026-09-14 10:12:31	10.0.0.19	DDoS	SYN Flood	91.0%	Critical	Time Ban (1m)
2026-09-14 10:12:31	10.0.0.19	DDoS	SYN Flood	91.0%	Critical	Time Ban
2026-09-14 10:13:26	10.0.0.16	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-14 10:13:26	10.0.0.18	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-14 10:13:27	10.0.0.17	DDoS	SYN Flood	91.0%	Critical	Released
2026-09-14 10:13:31	10.0.0.19	DDoS	SYN Flood	92.1%	Critical	Time Ban (2m)
2026-09-14 10:13:37	10.0.0.17	DDoS	SYN Flood	92.1%	Critical	Time Ban (2m)
2026-09-14 10:19:27	10.0.0.16	DDoS	SYN Flood	94.0%	Critical	Released
2026-09-14 10:19:28	10.0.0.17	DDoS	SYN Flood	93.3%	Critical	Blackhole
2026-09-14 10:19:38	10.0.0.16	DDoS	SYN Flood	94.0%	Critical	Blackhole
2026-09-14 10:23:42	10.0.0.18	DDoS	SYN Flood	91.0%	Critical	Blackhole
2026-09-14 10:23:45	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Blackhole
2026-09-14 10:29:13	10.0.0.5	DDoS	Uncertain	59.1%	Low	Released
2026-09-14 10:29:18	10.0.0.1	DDoS	Uncertain	58.9%	Low	Released
2026-09-14 10:29:18	10.0.0.8	DDoS	Uncertain	68.6%	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.5	Uncertain	0.5948	61.3%	Low	Phase 1	1	11s	2026-09-14 10:30:40	Attack Stopped
10.0.0.5	Uncertain	0.6109	61.2%	Low	Phase 1	1	11s	2026-09-14 10:29:36	Attack Stopped
10.0.0.1	Uncertain	0.6064	58.9%	Low	Phase 1	1	11s	2026-09-14 10:29:18	Attack Stopped
10.0.0.8	Uncertain	0.5704	68.6%	Low	Phase 1	1	11s	2026-09-14 10:29:18	Attack Stopped
10.0.0.5	Uncertain	0.5915	59.1%	Low	Phase 1	1	11s	2026-09-14 10:29:13	Attack Stopped

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6480	93.3%	Critical	Phase 3	1	0s	2026-09-14 10:28:48	Escalated to Blackhole
10.0.0.16	SYN Flood	0.6342	91.5%	Critical	Phase 3	1	1s	2026-09-14 10:28:45	Escalated to Blackhole
10.0.0.18	SYN Flood	0.6307	91.4%	Critical	Phase 3	1	0s	2026-09-14 10:28:45	Escalated to Blackhole
10.0.0.17	SYN Flood	0.6413	93.2%	Critical	Phase 3	1	0s	2026-09-14 10:28:45	Escalated to Blackhole
10.0.0.19	SYN Flood	0.6513	90.9%	Critical	Phase 2	1	10s	2026-09-14 10:23:45	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6514	90.9%	Critical	Phase 3	2	10s	2026-09-14 10:23:45	Escalated to Blackhole
10.0.0.18	SYN Flood	0.6464	91.0%	Critical	Phase 3	2	11s	2026-09-14 10:23:42	Escalated to Blackhole
10.0.0.18	SYN Flood	0.6462	91.0%	Critical	Phase 2	1	10s	2026-09-14 10:23:41	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6370	92.1%	Critical	Phase 3	1	1s	2026-09-14 10:23:32	Escalated to Blackhole
10.0.0.16	SYN Flood	0.6348	93.4%	Critical	Phase 3	1	0s	2026-09-14 10:23:30	Escalated to Blackhole
10.0.0.16	SYN Flood	0.6357	94.0%	Critical	Phase 2	3	11s	2026-09-14 10:19:38	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6406	94.0%	Critical	Phase 3	4	11s	2026-09-14 10:19:38	Escalated to Blackhole
10.0.0.19	SYN Flood	0.6476	93.2%	Critical	Phase 2	2	1m 11s	2026-09-14 10:19:31	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6297	93.1%	Critical	Phase 2	2	1m 11s	2026-09-14 10:19:28	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6330	93.3%	Critical	Phase 2	2	1m 11s	2026-09-14 10:19:28	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6327	93.3%	Critical	Phase 3	3	1m 11s	2026-09-14 10:19:28	Escalated to Blackhole
10.0.0.16	SYN Flood	0.5968	94.0%	Critical	Phase 2	1	1m 11s	2026-09-14 10:19:27	Ban Expired
10.0.0.19	SYN Flood	0.6475	91.9%	Critical	Phase 2	1	10s	2026-09-14 10:18:30	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6353	93.1%	Critical	Phase 2	1	10s	2026-09-14 10:18:27	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6425	93.3%	Critical	Phase 2	1	10s	2026-09-14 10:18:27	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.16	SYN Flood	0.6368	91.7%	Critical	Phase 2	1	10s	2026-09-14 10:18:26	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6299	92.1%	Critical	Phase 2	3	10s	2026-09-14 10:13:37	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6523	92.1%	Critical	Phase 2	2	1m 11s	2026-09-14 10:13:31	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6014	91.0%	Critical	Phase 2	1	1m 11s	2026-09-14 10:13:27	Ban Expired
10.0.0.16	SYN Flood	0.6275	93.4%	Critical	Phase 2	2	1m 11s	2026-09-14 10:13:26	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6368	94.0%	Critical	Phase 2	2	1m 10s	2026-09-14 10:13:26	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6524	91.0%	Critical	Phase 2	1	11s	2026-09-14 10:12:31	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6376	91.0%	Critical	Phase 2	1	11s	2026-09-14 10:12:27	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6376	93.4%	Critical	Phase 2	1	11s	2026-09-14 10:12:26	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6443	91.0%	Critical	Phase 2	1	10s	2026-09-14 10:12:26	Attack persisted - escalating to time ban
10.0.0.9	UDP Flood	0.6124	92.4%	High	Phase 1	3	10s	2026-09-14 10:09:59	Attack Stopped
10.0.0.9	Uncertain	0.5654	69.9%	Low	Phase 1	2	11s	2026-09-14 10:09:16	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE